

CSS Colors Guide

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1. Introduction to CSS Colors

CSS provides multiple ways to define colors, making web design more vibrant and visually appealing.

2. Ways to Define Colors in CSS

a) Named Colors

CSS supports 140+ predefined color names such as:

- **Red**
- **Blue**
- **Green**
- **Yellow**
- **Black** and **White**

Example:

```
```css
p {
 color: red;
}
```
```

b) HEX Colors

A HEX code is a 6-digit code representing colors.

- Format: `#RRGGBB``
- Example:

```
```css
h1 {
 color: #ff5733; /* Orange */
}
```
```

c) RGB (Red, Green, Blue)

Defines colors using RGB values, with each component ranging from 0-255.

- Example:

```
```css
div {
 background-color: rgb(0, 255, 0); /* Green */
}
```
```

d) RGBA (RGB + Alpha)

Similar to RGB but includes an **alpha channel** (transparency) ranging from 0 (fully transparent) to 1 (fully opaque).

- Example:

```
```css
button {
 background-color: rgba(255, 0, 0, 0.5); /* Semi-transparent red */
}
```
```

e) HSL (Hue, Saturation, Lightness)

Defines colors based on:

- **Hue**: 0° (Red) to 360° (full spectrum)
- **Saturation**: 0% (gray) to 100% (full color)
- **Lightness**: 0% (black) to 100% (white)

Example:

```
```css
span {
 color: hsl(200, 100%, 50%); /* Bright blue */
}
```
```

f) HSLA (HSL + Alpha)

Adds transparency to HSL colors.

- Example:

```
```css
div {
 background-color: hsla(120, 50%, 50%, 0.3); /* Transparent green */
}
```
```

...

3. CSS Gradient Colors

a) Linear Gradient

A transition between multiple colors in a straight line.

Example:

```
```css
background: linear-gradient(to right, red, blue);
```
```

b) Radial Gradient

A color transition that spreads from the center.

Example:

```
```css
background: radial-gradient(circle, yellow, green);
```
```

Conclusion

Using different color methods in CSS helps create visually appealing and accessible websites.

Experiment with various color combinations to enhance your designs!